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AIUB, Machine Learning Notes

MACHINE LEARNING

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ENSEMBLE LEARNING

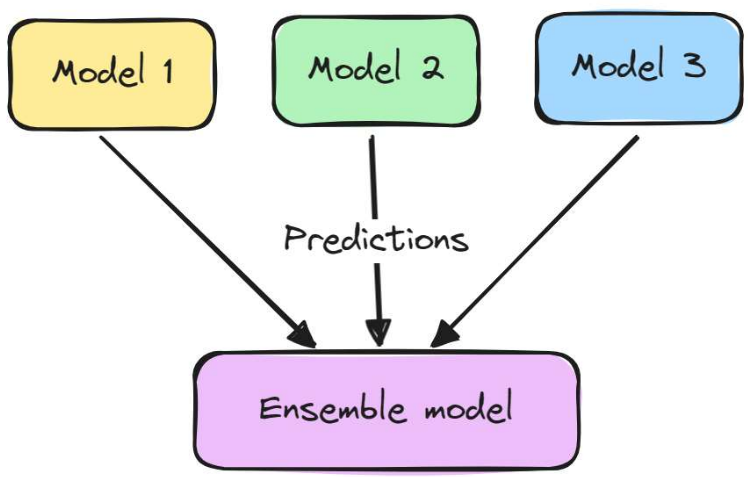
Ensemble Learning

⇨ Ensemble learning is a machine learning technique that combines multiple individual models to improve overall **performance, accuracy, and robustness**.

⇨ The main idea is that a group of **weak learners** (slightly better than random guessing) or **strong learners** can together produce a model that outperforms any single model.

❖ **Advantages:**

- Reduces overfitting
- Improves generalization
- Handles noisy data effectively



Types of Ensemble Learning Techniques

1. **Bagging (Bootstrap Aggregating)**
2. **Boosting**
3. **Stacking (Stacked Generalization)**

Bagging (Bootstrap Aggregating)

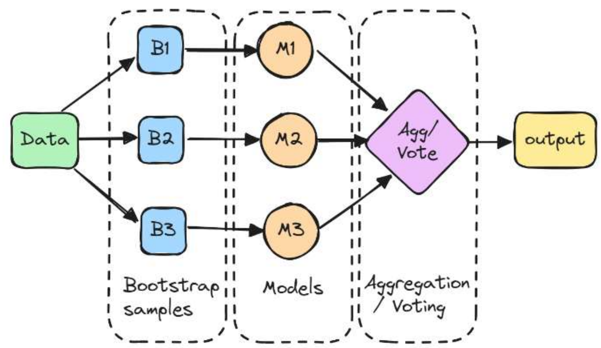
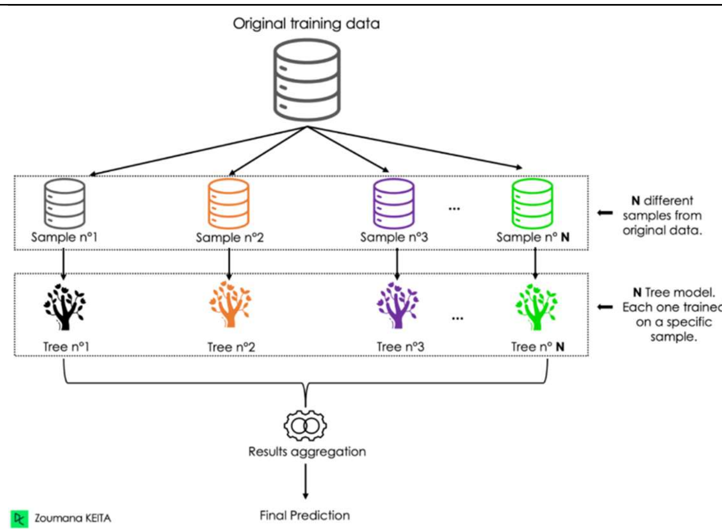
⇨ Bagging trains multiple models **in parallel** using different subsets of the training data created through **random sampling with replacement**.

❖ **Working Principle:**

- Create multiple bootstrap samples from the dataset
- Train a separate model on each sample
- Combine predictions using:
  - **Majority voting** (classification)
  - **Averaging** (regression)

❖ **Example:**

- **Random Forest:** Popular bagging algorithm that uses multiple decision trees.

Block Diagram:

Model Diagram:


Boosting

⇨ Boosting builds models **sequentially**, where each new model focuses on correcting the errors of the previous one.

❖ Key Characteristics:

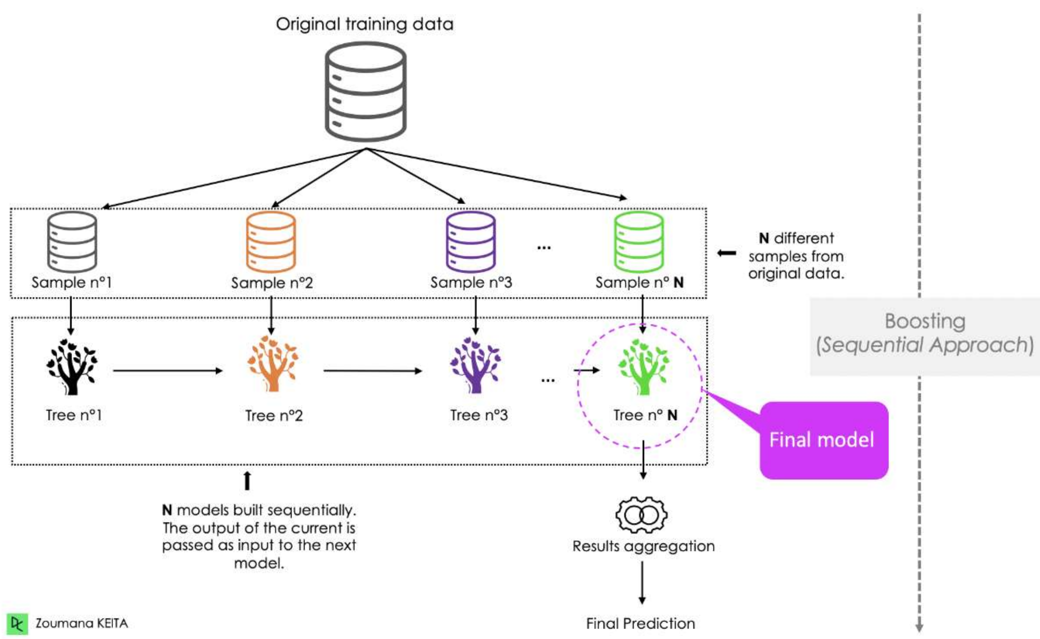
- Data points are assigned **weights**
- Misclassified instances receive higher importance
- Reduces **bias** and improves accuracy

❖ Working Principle:

- **Assign Initial Weights:** All data points are given equal importance initially
- **Sequential Training:** Train a weak learner and evaluate errors
- **Weight Adjustment:** Increase weights of misclassified samples
- **Repeat Process:** Train additional learners focusing on harder cases
- **Aggregate Results:** Combine all learners using weighted voting

❖ Common Boosting Algorithms:

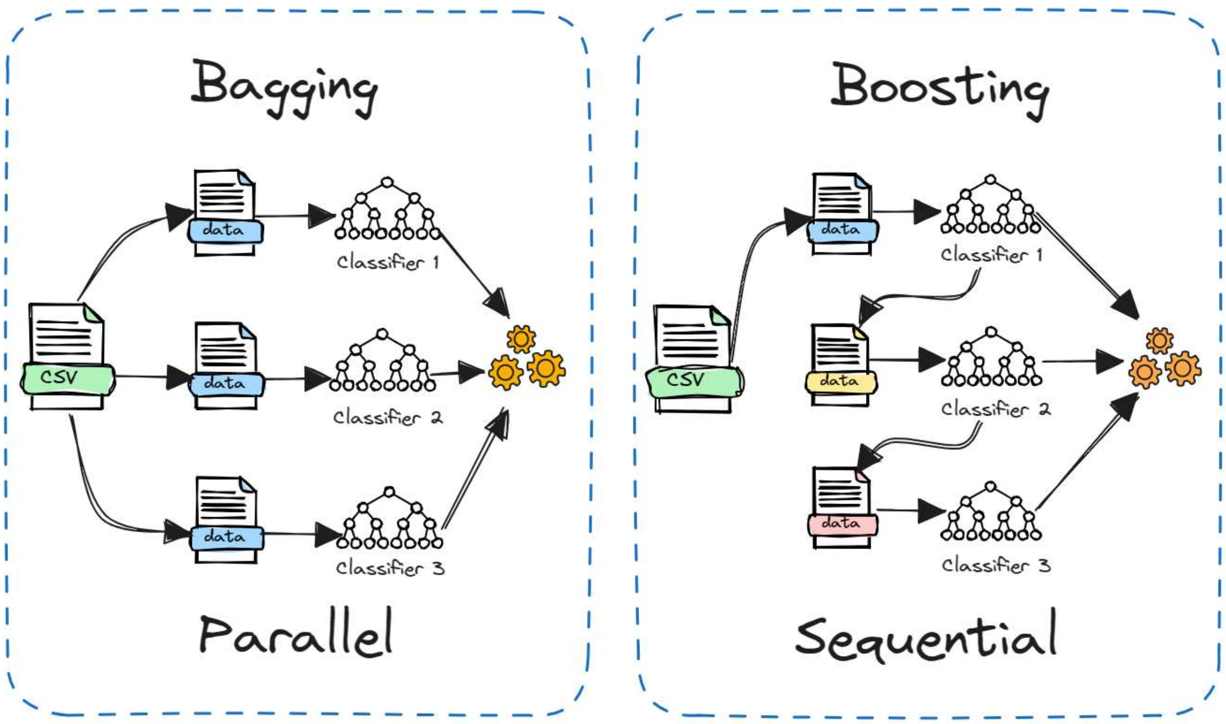
1. AdaBoost
2. Gradient Boosting
3. XGBoost



Bagging vs. Boosting

Aspect	Bagging	Boosting
Training	Parallel	Sequential
Focus	Reducing variance	Reducing bias
Data Sampling	Random with replacement	Reweighted samples
Model Combination	Simple averaging	Weighted aggregation

- **Block Diagram:**



Stacking (Stacked Generalization)

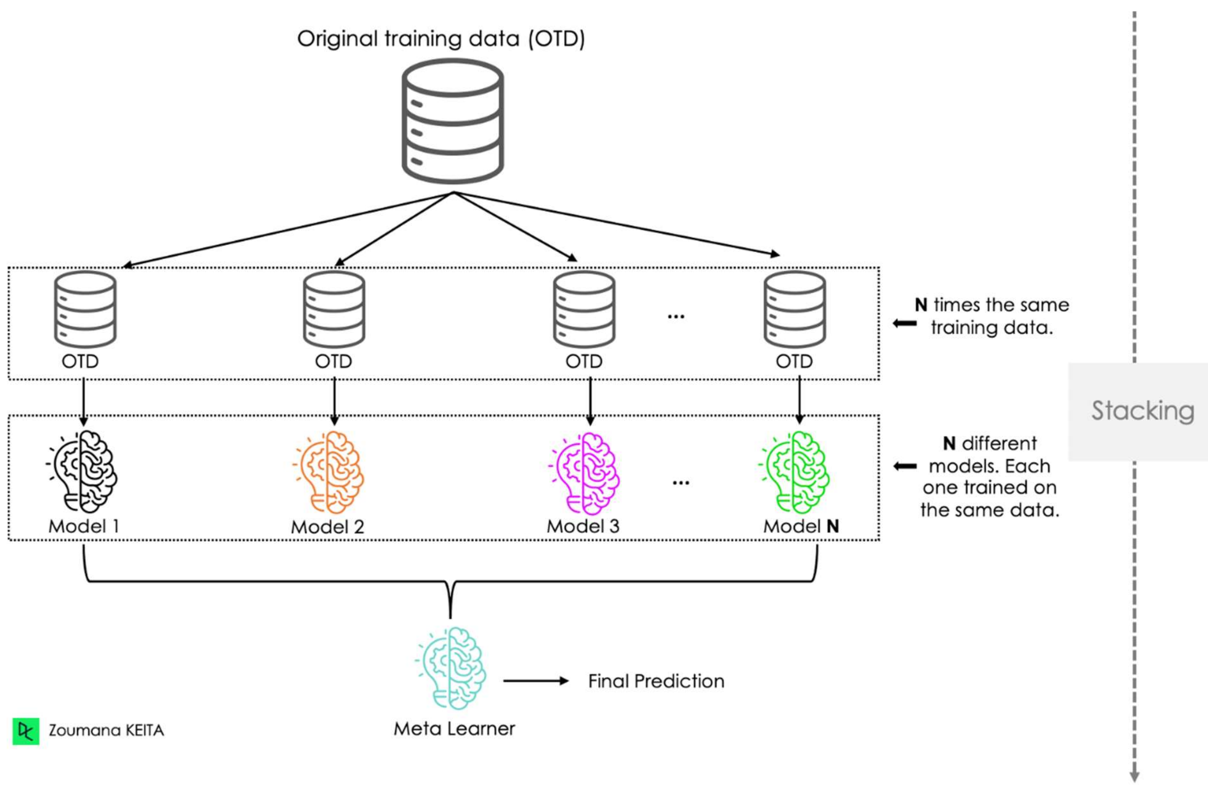
⇨ Stacking combines **different types of models** trained on the same dataset. A **meta-model** is then trained to learn how to best combine their predictions.

❖ **Key Points:**

- Allows heterogeneous models
- Captures diverse patterns
- Meta-model learns optimal combination strategy

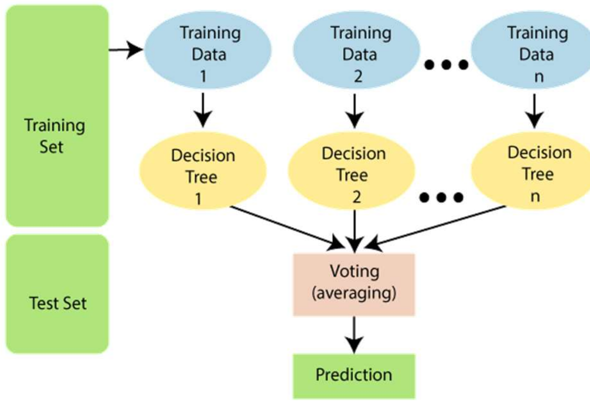
❖ **Example:**

1. **Base models:** Decision Tree, Logistic Regression, SVM
2. **Meta-model:** Linear Regression



Random Forest

⇨ Random Forest is an ensemble algorithm mainly used for **classification and regression**. It is based on the **bagging** technique and uses multiple decision trees.



Key Features of Random Forest

❖ Random Subsampling of Data:

- Each tree is trained on a different bootstrap sample
- Ensures diversity among trees

❖ Feature Randomness:

- At each split, only a random subset of features is considered
- Reduces correlation between trees
- Prevents overfitting

❖ Prediction Aggregation:

- **Classification:** Majority voting
- **Regression:** Averaging predictions

Problem1: You are given the following dataset used to predict whether a customer **buys a computer**.

ID	Age	Income	Student	Credit_Rating	Buys_Computer
1	≤30	High	No	Fair	No
2	≤30	High	No	Excellent	No
3	31– 40	High	No	Fair	Yes
4	>40	Medium	No	Fair	Yes
5	>40	Low	Yes	Fair	Yes
6	>40	Low	Yes	Excellent	No

a) **Create** four bootstrap samples**, each of size** 6**, from the dataset using** sampling with replacement**.**

b) **Create** First three decision tree classifiers**, one on each bootstrap sample**.

c) For a new customer with the following attributes:

Age = >40
Income = Low
Student = Yes
Credit_Rating = Fair

Using **bagging** determine the **final predicted class** for *Buys_Computer*.

(a)

Sample1: ID: 1,1,3,4,5,6

ID	Age	Income	Student	Credit Rating	Buys Computer
1	≤30	High	No	Fair	No
1	≤30	High	No	Fair	No
3	31– 40	High	No	Fair	Yes
4	>40	Medium	No	Fair	Yes
5	>40	Low	Yes	Fair	Yes
6	>40	Low	Yes	Excellent	No

Sample2: ID: 1,2,3,4,2,6

ID	Age	Income	Student	Credit Rating	Buys Computer
1	≤30	High	No	Fair	No
2	≤30	High	No	Excellent	No
3	31– 40	High	No	Fair	Yes
4	>40	Medium	No	Fair	Yes
2	≤30	High	No	Excellent	No
6	>40	Low	Yes	Excellent	No

Sample3: ID: 2,1,3,4,5,3

ID	Age	Income	Student	Credit Rating	Buys Computer
2	≤30	High	No	Excellent	No
1	≤30	High	No	Fair	No
3	31– 40	High	No	Fair	Yes
4	>40	Medium	No	Fair	Yes
5	>40	Low	Yes	Fair	Yes
3	31– 40	High	No	Fair	Yes

Sample4: ID: 1,3,5,4,5,6

ID	Age	Income	Student	Credit Rating	Buys Computer
1	≤30	High	No	Fair	No
3	31– 40	High	No	Fair	Yes
5	>40	Low	Yes	Fair	Yes
4	>40	Medium	No	Fair	Yes
5	>40	Low	Yes	Fair	Yes
6	>40	Low	Yes	Excellent	No

(b)

⇨ **For Bootstrap Sample1:** ID: 1,1,3,4,5,6

We Know,

$$\boldsymbol{Entropy}(\boldsymbol{D}) = -\boldsymbol{p_1 log_2}(\boldsymbol{p_1}) - \boldsymbol{p_2 log_2}(\boldsymbol{p_2})$$

$$\boldsymbol{Entropy}(\boldsymbol{D}) = -\frac{3}{6} log_2\left(\frac{3}{6}\right) - \frac{3}{6} log_2\left(\frac{3}{6}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D}) = 1$$

∴ **Entropy of the entire dataset** for the target attribute ***buys_computer***, $\boldsymbol{D} = 1$

❖ **For Age** (values: <=30, 31..40, >40):

Is given, For Age $\leq$ **30**:

$$\boldsymbol{Entropy}(\boldsymbol{D_{v\leq 30}}) = -\frac{2}{2} log_2\left(\frac{2}{2}\right) - \frac{0}{2} log_2\left(\frac{0}{2}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D_{v\leq 30}}) = 0$$

Is given, For Age **31 ... 40**:

$$\boldsymbol{Entropy}(\boldsymbol{D_{v\,31..40}}) = -\frac{1}{1} log_2\left(\frac{1}{1}\right) - \frac{0}{1} log_2\left(\frac{0}{1}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D_{v\,31..40}}) = 0$$

Is given, For Age $>$ **40**:

$$\boldsymbol{Entropy}(\boldsymbol{D_{v>40}}) = -\frac{2}{3} log_2\left(\frac{2}{3}\right) - \frac{1}{3} log_2\left(\frac{1}{3}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D_{v>40}}) = 0.918$$

So,

$$\therefore \mathbf{IG}(\mathbf{D}, \mathbf{Age}) = \mathbf{Entropy}(\mathbf{D}) - \sum \frac{|\boldsymbol{D_v}|}{|\boldsymbol{D}|} \times \mathbf{Entropy}(\boldsymbol{D_v})$$

$$\boxed{\mathbf{IG}(\mathbf{D}, \mathbf{Age}) = 1 - \left(\frac{|2|}{|6|} \times \mathbf{0} + \frac{|1|}{|6|} \times \mathbf{0} + \frac{|3|}{|6|} \times \mathbf{0.918}\right) = 0.541}$$

❖ **For Income** (values: high, medium, low):

Is given, For income, High:

$$\boldsymbol{Entropy}(\boldsymbol{D}_{\boldsymbol{v}\ \boldsymbol{high}}) = -\frac{2}{3}log_2\left(\frac{2}{3}\right) - \frac{1}{3}log_2\left(\frac{1}{3}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D}_{\boldsymbol{v}\ \boldsymbol{high}}) = 0.918$$

Is given, For income, Medium:

$$\boldsymbol{Entropy}(\boldsymbol{D}_{\boldsymbol{v}\ \boldsymbol{medium}}) = -\frac{1}{1}log_2\left(\frac{1}{1}\right) - \frac{0}{1}log_2\left(\frac{0}{1}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D}_{\boldsymbol{v}\ \boldsymbol{medium}}) = 0$$

Is given, For income, Low:

$$\boldsymbol{Entropy}(\boldsymbol{D}_{\boldsymbol{v}\ \boldsymbol{low}}) = -\frac{1}{2}log_2\left(\frac{1}{2}\right) - \frac{1}{2}log_2\left(\frac{1}{2}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D}_{\boldsymbol{v}\ \boldsymbol{low}}) = 1$$

So,

$$\therefore \mathbf{IG}(\mathbf{D}, \mathbf{Income}) = \mathbf{Entropy}(\mathbf{D}) - \sum \frac{|\boldsymbol{D_v}|}{|\boldsymbol{D}|} \times \mathbf{Entropy}(\boldsymbol{D_v})$$

$$\boxed{\mathbf{IG}(\mathbf{D}, \mathbf{Income}) = 1 - \left(\frac{|3|}{|6|} \times \mathbf{0.918} + \frac{|1|}{|6|} \times \mathbf{0} + \frac{|2|}{|6|} \times \mathbf{1}\right) = 0.208}$$

❖ **For Student** (values: yes, no):

Is given, For Student, yes:

$$\boldsymbol{Entropy}(\boldsymbol{D}_{\boldsymbol{v}\ \boldsymbol{yes}}) = -\frac{2}{4}log_2\left(\frac{2}{4}\right) - \frac{2}{4}log_2\left(\frac{2}{4}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D}_{\boldsymbol{v}\ \boldsymbol{yes}}) = 1$$

Is given, For Student, No:

$$\boldsymbol{Entropy}(\boldsymbol{D}_{\boldsymbol{v}\ \boldsymbol{no}}) = -\frac{1}{2}log_2\left(\frac{1}{2}\right) - \frac{1}{2}log_2\left(\frac{1}{2}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D}_{\boldsymbol{v}\ \boldsymbol{no}}) = 1$$

So,

$$\therefore \mathbf{IG}(\mathbf{D}, \mathbf{Student}) = \mathbf{Entropy}(\mathbf{D}) - \sum \frac{|\boldsymbol{D_v}|}{|\boldsymbol{D}|} \times \mathbf{Entropy}(\boldsymbol{D_v})$$

$$\boxed{\mathbf{IG}(\mathbf{D}, \mathbf{Student}) = 1 - \left(\frac{|4|}{|6|} \times \mathbf{1} + \frac{|2|}{|6|} \times \mathbf{1}\right) = 0}$$

❖ **For credit_rating** (values: fair, excellent):

Is given, For credit_rating, fair:

$$\boldsymbol{Entropy}(\boldsymbol{D}_{v\,fair}) = -\frac{3}{5}log_2\left(\frac{3}{5}\right) - \frac{2}{5}log_2\left(\frac{2}{5}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D}_{v\,fair}) = 0.971$$

Is given, For credit_rating, excellent:

$$\boldsymbol{Entropy}(\boldsymbol{D}_{v\,excellent}) = -\frac{1}{1}log_2\left(\frac{1}{1}\right) - \frac{0}{1}log_2\left(\frac{0}{1}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D}_{v\,excellent}) = 0$$

So,

$$\therefore \mathbf{IG(D, credit\_rating)} = \mathbf{Entropy(D)} - \sum \frac{|\boldsymbol{D_v}|}{|\boldsymbol{D}|} \times \mathbf{Entropy}(\boldsymbol{D_v})$$

$$\boxed{\mathbf{IG(D, credit\_rating)} = 1 - \left(\frac{|5|}{|6|} \times \mathbf{0.971} + \frac{|1|}{|6|} \times \mathbf{0}\right) = 0.191}$$

We have,

$$\mathbf{IG(D, Age)} = 0.541$$

$$\mathbf{IG(D, Income)} = 0.208$$

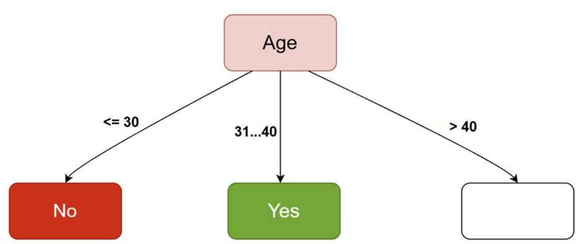
$$\mathbf{IG(D, Student)} = 0.000$$

$$\mathbf{IG(D, credit\_rating\ )} = 0.191$$

We Know,

$$\boldsymbol{Root\ attribut = Attribute\ with\ Highest\ Information\ Gain}$$

$$\boldsymbol{Root\ attribut = age}$$



Age	Income	Student	Credit Rating	Buys Computer
>40	Medium	No	Fair	Yes
>40	Low	Yes	Fair	Yes
>40	Low	Yes	Excellent	No

We Know,

$$\boldsymbol{Entropy(D)} = -\boldsymbol{p_1 log_2(p_1)} - \boldsymbol{p_2 log_2(p_2)}$$

$$\boldsymbol{Entropy(D)} = -\frac{2}{3}log_2\left(\frac{2}{3}\right) - \frac{1}{3}log_2\left(\frac{1}{3}\right)$$

$$\boldsymbol{Entropy(D)} = 0.918$$

$\therefore$ **Entropy of the entire dataset** for the target attribute ***buys_computer***, $\boldsymbol{D} = 0.918$

- **For Income** (values: high, medium, low):

  Is given, For income, High:

$$\boldsymbol{Entropy(D_{v\ high})} = -\frac{0}{0}log_2\left(\frac{0}{0}\right) - \frac{0}{0}log_2\left(\frac{0}{0}\right)$$

$$\boldsymbol{Entropy(D_{v\ high})} = 0$$

  Is given, For income, Medium:

$$\boldsymbol{Entropy(D_{v\ medium})} = -\frac{1}{1}log_2\left(\frac{1}{1}\right) - \frac{0}{1}log_2\left(\frac{0}{1}\right)$$

$$\boldsymbol{Entropy(D_{v\ medium})} = 0$$

  Is given, For income, Low:

$$\boldsymbol{Entropy(D_{v\ low})} = -\frac{1}{2}log_2\left(\frac{1}{2}\right) - \frac{1}{2}log_2\left(\frac{1}{2}\right)$$

$$\boldsymbol{Entropy(D_{v\ low})} = 1$$

So,

$$\therefore \mathbf{IG(D, Income\ )} = \mathbf{Entropy(D)} - \sum \frac{|\boldsymbol{D_v}|}{|\boldsymbol{D}|} \times \mathbf{Entropy}(\boldsymbol{D_v})$$

$$\boxed{\mathbf{IG(D, Income\ )} = 0.918 - \left(\frac{|0|}{|3|} \times \mathbf{0} + \frac{|1|}{|3|} \times \mathbf{0} + \frac{|2|}{|3|} \times \mathbf{1}\right) = 0.251}$$

- **For Student** (values: yes, no):

  Is given, For Student, yes:

$$\boldsymbol{Entropy(D_{v\ yes})} = -\frac{1}{2}log_2\left(\frac{1}{2}\right) - \frac{1}{2}log_2\left(\frac{1}{2}\right)$$

$$\boldsymbol{Entropy(D_{v\ yes})} = 1$$

  Is given, For Student, No:

$$\boldsymbol{Entropy(D_{v\ no})} = -\frac{1}{1}log_2\left(\frac{1}{1}\right) - \frac{0}{1}log_2\left(\frac{0}{1}\right)$$

$$\boldsymbol{Entropy(D_{v\ no})} = 0$$

So,

$$\therefore \mathbf{IG(D, Student\ )} = \mathbf{Entropy(D)} - \sum \frac{|D_v|}{|D|} \times \mathbf{Entropy}(\boldsymbol{D_v})$$

$$\boxed{\mathbf{IG(D, Student\ )} = 0.918 - \left(\frac{|2|}{|3|} \times \mathbf{1} + \frac{|1|}{|3|} \times \mathbf{0}\right) = 0.251}$$

❖ **For credit_rating** (values: fair, excellent):

Is given, For credit_rating, fair:

$$\boldsymbol{Entropy}(\boldsymbol{D}_{\boldsymbol{v}\ \boldsymbol{fair}}) = -\frac{2}{2} log_2\left(\frac{2}{2}\right) - \frac{0}{2} log_2\left(\frac{0}{2}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D}_{\boldsymbol{v}\ \boldsymbol{fair}}) = 0$$

Is given, For credit_rating, excellent:

$$\boldsymbol{Entropy}(\boldsymbol{D}_{\boldsymbol{v}\ \boldsymbol{excellent}}) = -\frac{1}{1} log_2\left(\frac{1}{1}\right) - \frac{0}{1} log_2\left(\frac{0}{1}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D}_{\boldsymbol{v}\ \boldsymbol{excellent}}) = 0$$

So,

$$\therefore \mathbf{IG(D, credit\_rating)} = \mathbf{Entropy(D)} - \sum \frac{|D_v|}{|D|} \times \mathbf{Entropy}(\boldsymbol{D_v})$$

$$\boxed{\mathbf{IG(D, credit\_rating)} = 0.918 - \left(\frac{|2|}{|3|} \times \mathbf{0} + \frac{|1|}{|3|} \times \mathbf{0}\right) = 0.918}$$

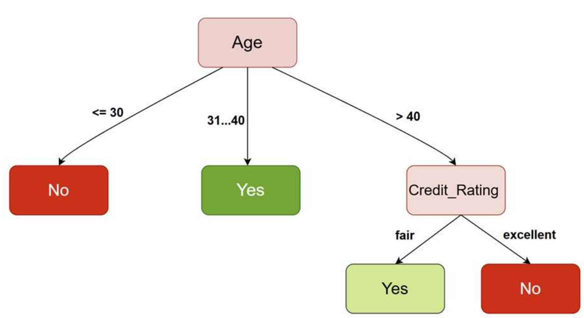
We have,

$$\mathbf{IG(D, Income)} = 0.251$$
$$\mathbf{IG(D, Student)} = 0.251$$
$$\mathbf{IG(D, credit\_rating\ )} = 0.918$$

We Know,

$$\boldsymbol{Root\ attribut} = \boldsymbol{Attribute\ with\ Highest\ Information\ Gain}$$

$$\boldsymbol{Root\ attribut} = \mathbf{credit\_rating}$$



⇨ **For Bootstrap Sample2:** ID: 1,2,3,4,2,6

We Know,

$$\boldsymbol{Entropy(D) = -p_1 log_2(p_1) - p_2 log_2(p_2)}$$

$$\boldsymbol{Entropy(D)} = -\frac{4}{6} log_2\left(\frac{4}{6}\right) - \frac{2}{6} log_2\left(\frac{2}{6}\right)$$

$$\boldsymbol{Entropy(D)} = 0.918$$

∴ **Entropy of the entire dataset** for the target attribute ***buys_computer***, $\boldsymbol{D} = 0.918$

❖ **For Age** (values: <=30, 31..40, >40):

Is given, For Age $\leq$ **30**:

$$\boldsymbol{Entropy(D_{v\leq 30})} = -\frac{3}{3} log_2\left(\frac{3}{3}\right) - \frac{0}{3} log_2\left(\frac{0}{3}\right)$$

$$\boldsymbol{Entropy(D_{v\leq 30})} = 0$$

Is given, For Age **31 ... 40**:

$$\boldsymbol{Entropy(D_{v\,31..40})} = -\frac{1}{1} log_2\left(\frac{1}{1}\right) - \frac{0}{1} log_2\left(\frac{0}{1}\right)$$

$$\boldsymbol{Entropy(D_{v\,31..40})} = 0$$

Is given, For Age $>$ **40**:

$$\boldsymbol{Entropy(D_{v>40})} = -\frac{1}{2} log_2\left(\frac{1}{2}\right) - \frac{1}{2} log_2\left(\frac{1}{2}\right)$$

$$\boldsymbol{Entropy(D_{v>40})} = 1$$

So,

$$\therefore \mathbf{IG(D, Age) = Entropy(D)} - \sum \frac{|\boldsymbol{D_v}|}{|\boldsymbol{D}|} \times \mathbf{Entropy}(\boldsymbol{D_v})$$

$$\boxed{\mathbf{IG(D, Age)} = 0.918 - \left(\frac{|3|}{|6|} \times \mathbf{0} + \frac{|1|}{|6|} \times \mathbf{0} + \frac{|2|}{|6|} \times \mathbf{1}\right) = 0.584}$$

❖ **For Income** (values: high, medium, low):

Is given, For income, High:

$$\boldsymbol{Entropy}(\boldsymbol{D}_{v\ high}) = -\frac{3}{4}log_2\left(\frac{3}{4}\right) - \frac{1}{4}log_2\left(\frac{1}{4}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D}_{v\ high}) = 0.811$$

Is given, For income, Medium:

$$\boldsymbol{Entropy}(\boldsymbol{D}_{v\ medium}) = -\frac{1}{1}log_2\left(\frac{1}{1}\right) - \frac{0}{1}log_2\left(\frac{0}{1}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D}_{v\ medium}) = 0$$

Is given, For income, Low:

$$\boldsymbol{Entropy}(\boldsymbol{D}_{v\ low}) = -\frac{1}{1}log_2\left(\frac{1}{1}\right) - \frac{0}{1}log_2\left(\frac{0}{1}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D}_{v\ low}) = 0$$

So,

$$\therefore \mathbf{IG}(\mathbf{D}, \mathbf{Income}) = \mathbf{Entropy}(\mathbf{D}) - \sum \frac{|\boldsymbol{D_v}|}{|\boldsymbol{D}|} \times \mathbf{Entropy}(\boldsymbol{D_v})$$

$$\boxed{\mathbf{IG}(\mathbf{D}, \mathbf{Income}) = 0.918 - \left(\frac{|4|}{|6|} \times \mathbf{0.811} + \frac{|1|}{|6|} \times \mathbf{0} + \frac{|1|}{|6|} \times \mathbf{0}\right) = 0.377}$$

❖ **For Student** (values: yes, no):

Is given, For Student, yes:

$$\boldsymbol{Entropy}(\boldsymbol{D}_{v\ yes}) = -\frac{1}{1}log_2\left(\frac{1}{1}\right) - \frac{0}{1}log_2\left(\frac{0}{1}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D}_{v\ yes}) = \mathbf{0}$$

Is given, For Student, No:

$$\boldsymbol{Entropy}(\boldsymbol{D}_{v\ no}) = -\frac{3}{5}log_2\left(\frac{3}{5}\right) - \frac{2}{5}log_2\left(\frac{2}{5}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D}_{v\ no}) = \mathbf{0}.970$$

So,

$$\therefore \mathbf{IG}(\mathbf{D}, \mathbf{Student}) = \mathbf{Entropy}(\mathbf{D}) - \sum \frac{|\boldsymbol{D_v}|}{|\boldsymbol{D}|} \times \mathbf{Entropy}(\boldsymbol{D_v})$$

$$\boxed{\mathbf{IG}(\mathbf{D}, \mathbf{Student}) = 0.918 - \left(\frac{|5|}{|6|} \times \mathbf{0.97} + \frac{|1|}{|6|} \times \mathbf{0}\right) = 0.109}$$

❖ **For credit_rating** (values: fair, excellent):

Is given, For credit_rating, fair:

$$Entropy(D_{v\,fair}) = -\frac{2}{3}log_2\left(\frac{2}{3}\right) - \frac{1}{3}log_2\left(\frac{1}{3}\right)$$

$$Entropy(D_{v\,fair}) = 0.918$$

Is given, For credit_rating, excellent:

$$Entropy(D_{v\,excellent}) = -\frac{3}{3}log_2\left(\frac{3}{3}\right) - \frac{0}{3}log_2\left(\frac{0}{3}\right)$$

$$Entropy(D_{v\,excellent}) = 0$$

So,

$$\therefore \mathbf{IG(D, credit\_rating)} = \mathbf{Entropy(D)} - \sum \frac{|D_v|}{|D|} \times \mathbf{Entropy}(D_v)$$

$$\boxed{\mathbf{IG(D, credit\_rating)} = 0.918 - \left(\frac{|3|}{|6|} \times \mathbf{0.918} + \frac{|3|}{|6|} \times \mathbf{0}\right) = 0.459}$$

We have,

$$\mathbf{IG(D, Age)} = 0.584$$

$$\mathbf{IG(D, Income)} = 0.377$$

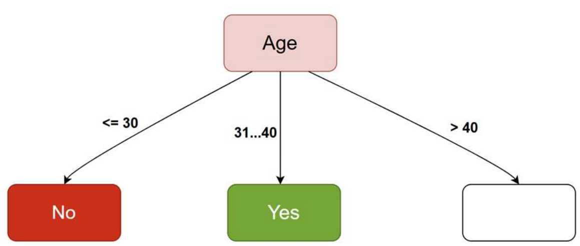
$$\mathbf{IG(D, Student)} = 0.109$$

$$\mathbf{IG(D, credit\_rating\ )} = 0.459$$

We Know,

$$Root\ attribut = Attribute\ with\ Highest\ Information\ Gain$$

$$Root\ attribut = age$$



ID	Age	Income	Student	Credit Rating	Buys Computer
4	>40	Medium	No	Fair	Yes
6	>40	Low	Yes	Excellent	No

We Know,

$$\boldsymbol{Entropy(D) = -p_1 log_2(p_1) - p_2 log_2(p_2)}$$

$$\boldsymbol{Entropy(D)} = -\frac{1}{2}log_2\left(\frac{1}{2}\right) - \frac{1}{2}log_2\left(\frac{1}{2}\right)$$

$$\boldsymbol{Entropy(D)} = 1$$

∴ **Entropy of the entire dataset** for the target attribute ***buys_computer***, $\boldsymbol{D} = 1$

❖ **For Income** (values: high, medium, low):

Is given, For income, High:

$$\boldsymbol{Entropy(D_{v\,high})} = -\frac{0}{0}log_2\left(\frac{2}{3}\right) - \frac{0}{0}log_2\left(\frac{0}{0}\right)$$

$$\boldsymbol{Entropy(D_{v\,high})} = 0$$

Is given, For income, Medium:

$$\boldsymbol{Entropy(D_{v\,medium})} = -\frac{1}{1}log_2\left(\frac{1}{1}\right) - \frac{0}{1}log_2\left(\frac{0}{1}\right)$$

$$\boldsymbol{Entropy(D_{v\,medium})} = 0$$

Is given, For income, Low:

$$\boldsymbol{Entropy(D_{v\,low})} = -\frac{1}{1}log_2\left(\frac{1}{1}\right) - \frac{0}{1}log_2\left(\frac{01}{1}\right)$$

$$\boldsymbol{Entropy(D_{v\,low})} = 0$$

So,

$$\therefore \textbf{IG}(\textbf{D}, \textbf{Income}) = \textbf{Entropy}(\textbf{D}) - \sum \frac{|\boldsymbol{D_v}|}{|\boldsymbol{D}|} \times \textbf{Entropy}(\boldsymbol{D_v})$$

$$\boxed{\textbf{IG}(\textbf{D}, \textbf{Income}) = 1 - \left(\frac{|\mathbf{0}|}{|\mathbf{2}|} \times \mathbf{0} + \frac{|\mathbf{1}|}{|\mathbf{2}|} \times \mathbf{0} + \frac{|\mathbf{1}|}{|\mathbf{2}|} \times \mathbf{0}\right) = 1}$$

❖ **For Student** (values: yes, no):

Is given, For Student, yes:

$$\boldsymbol{Entropy(D_{v\,yes})} = -\frac{1}{1}log_2\left(\frac{1}{1}\right) - \frac{0}{1}log_2\left(\frac{0}{1}\right)$$

$$\boldsymbol{Entropy(D_{v\,yes})} = 0$$

Is given, For Student, No:

$$\boldsymbol{Entropy(D_{v\,no})} = -\frac{1}{1}log_2\left(\frac{1}{1}\right) - \frac{0}{1}log_2\left(\frac{0}{1}\right)$$

$$\boldsymbol{Entropy(D_{v\,no})} = 0$$

So,

$$\therefore \mathbf{IG(D, Student)} = \mathbf{Entropy(D)} - \sum \frac{|D_v|}{|D|} \times \mathbf{Entropy}(D_v)$$

$$\boxed{\mathbf{IG(D, Student)} = 1 - \left(\frac{|1|}{|2|} \times \mathbf{0} + \frac{|1|}{|2|} \times \mathbf{0}\right) = 1}$$

❖ **For credit_rating** (values: fair, excellent):

Is given, For credit_rating, fair:

$$Entropy(D_{v\ fair}) = -\frac{1}{1}log_2\left(\frac{1}{1}\right) - \frac{0}{1}log_2\left(\frac{0}{1}\right)$$

$$Entropy(D_{v\ fair}) = 0$$

Is given, For credit_rating, excellent:

$$Entropy(D_{v\ excellent}) = -\frac{1}{1}log_2\left(\frac{1}{1}\right) - \frac{0}{1}log_2\left(\frac{0}{1}\right)$$

$$Entropy(D_{v\ excellent}) = 0$$

So,

$$\therefore \mathbf{IG(D, credit\_rating)} = \mathbf{Entropy(D)} - \sum \frac{|D_v|}{|D|} \times \mathbf{Entropy}(D_v)$$

$$\boxed{\mathbf{IG(D, credit\_rating)} = 1 - \left(\frac{|1|}{|2|} \times \mathbf{0} + \frac{|1|}{|2|} \times \mathbf{0}\right) = 1}$$

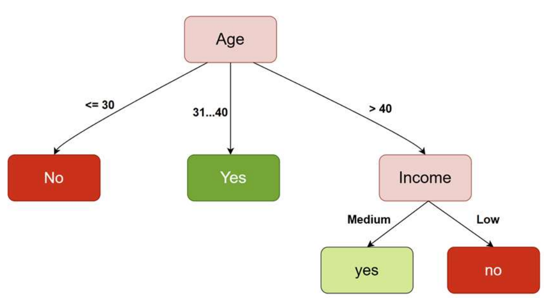
We have,

$$\mathbf{IG(D, Income)} = 1$$
$$\mathbf{IG(D, Student)} = 1$$
$$\mathbf{IG(D, credit\_rating)} = 1$$

We Know,

$$Root\ attribut = Attribute\ with\ Highest\ Information\ Gain$$

$$Root\ attribut = \mathbf{Income}$$



⇨ **For Bootstrap Sample3:** ID: 2,1,3,4,5,3

We Know,

$$\boldsymbol{Entropy(D) = -p_1 log_2(p_1) - p_2 log_2(p_2)}$$

$$\boldsymbol{Entropy(D)} = -\frac{4}{6}log_2\left(\frac{4}{6}\right) - \frac{2}{6}log_2\left(\frac{2}{6}\right)$$

$$\boldsymbol{Entropy(D)} = 0.918$$

∴ **Entropy of the entire dataset** for the target attribute ***buys_computer***, $\boldsymbol{D} = 0.918$

❖ **For Age** (values: <=30, 31..40, >40):

Is given, For Age $\leq$ **30**:

$$\boldsymbol{Entropy(D_{v\leq30})} = -\frac{2}{2}log_2\left(\frac{2}{2}\right) - \frac{0}{2}log_2\left(\frac{0}{2}\right)$$

$$\boldsymbol{Entropy(D_{v\leq30})} = 0$$

Is given, For Age **31 ... 40**:

$$\boldsymbol{Entropy(D_{v\,31..40})} = -\frac{2}{2}log_2\left(\frac{2}{2}\right) - \frac{0}{2}log_2\left(\frac{0}{2}\right)$$

$$\boldsymbol{Entropy(D_{v\,31..40})} = 0$$

Is given, For Age $>$ **40**:

$$\boldsymbol{Entropy(D_{v>40})} = -\frac{2}{2}log_2\left(\frac{2}{2}\right) - \frac{0}{2}log_2\left(\frac{0}{2}\right)$$

$$\boldsymbol{Entropy(D_{v>40})} = 0$$

So,

$$\therefore \mathbf{IG(D, Age) = Entropy(D)} - \sum \frac{|D_v|}{|D|} \times \mathbf{Entropy}(\boldsymbol{D_v})$$

$$\boxed{\mathbf{IG(D, Age)} = 0.918 - \left(\frac{|2|}{|6|} \times \mathbf{0} + \frac{|2|}{|6|} \times \mathbf{0} + \frac{|2|}{|6|} \times \mathbf{0}\right) = 0.918}$$

- **For Income** (values: high, medium, low):

Is given, For income, High:

$$Entropy(D_{v\ high}) = -\frac{2}{4}log_2\left(\frac{2}{4}\right) - \frac{2}{4}log_2\left(\frac{2}{4}\right)$$

$$Entropy(D_{v\ high}) = 1$$

Is given, For income, Medium:

$$Entropy(D_{v\ medium}) = -\frac{1}{1}log_2\left(\frac{1}{1}\right) - \frac{0}{1}log_2\left(\frac{0}{1}\right)$$

$$Entropy(D_{v\ medium}) = 0$$

Is given, For income, Low:

$$Entropy(D_{v\ low}) = -\frac{1}{1}log_2\left(\frac{1}{1}\right) - \frac{0}{1}log_2\left(\frac{0}{1}\right)$$

$$Entropy(D_{v\ low}) = 0$$

So,

$$\therefore \mathbf{IG(D, Income\ ) = Entropy(D) - \sum \frac{|D_v|}{|D|} \times Entropy(D_v)}$$

$$\boxed{\mathbf{IG(D, Income\ )} = 0.918 - \left(\frac{|4|}{|6|} \times \mathbf{1} + \frac{|1|}{|6|} \times \mathbf{0} + \frac{|1|}{|6|} \times \mathbf{0}\right) = 0.251}$$

- **For Student** (values: yes, no):

Is given, For Student, yes:

$$Entropy(D_{v\ yes}) = -\frac{1}{1}log_2\left(\frac{1}{1}\right) - \frac{0}{1}log_2\left(\frac{0}{1}\right)$$

$$Entropy(D_{v\ yes}) = \mathbf{0}$$

Is given, For Student, No:

$$Entropy(D_{v\ no}) = -\frac{3}{5}log_2\left(\frac{3}{5}\right) - \frac{2}{5}log_2\left(\frac{2}{5}\right)$$

$$Entropy(D_{v\ no}) = \mathbf{0.}970$$

So,

$$\therefore \mathbf{IG(D, Student\ ) = Entropy(D) - \sum \frac{|D_v|}{|D|} \times Entropy(D_v)}$$

$$\boxed{\mathbf{IG(D, Student\ )} = 0.918 - \left(\frac{|5|}{|6|} \times \mathbf{0.97} + \frac{|1|}{|6|} \times \mathbf{0}\right) = 0.109}$$

❖ **For credit_rating** (values: fair, excellent):

Is given, For credit_rating, fair:

$$\boldsymbol{Entropy}(\boldsymbol{D}_{v\ fair}) = -\frac{4}{5}log_2\left(\frac{4}{5}\right) - \frac{1}{5}log_2\left(\frac{1}{5}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D}_{v\ fair}) = 0.721$$

Is given, For credit_rating, excellent:

$$\boldsymbol{Entropy}(\boldsymbol{D}_{v\ excellent}) = -\frac{1}{1}log_2\left(\frac{1}{1}\right) - \frac{0}{1}log_2\left(\frac{0}{1}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D}_{v\ excellent}) = 0$$

So,

$$\therefore \mathbf{IG}(\mathbf{D}, \mathbf{credit\_rating}) = \mathbf{Entropy}(\mathbf{D}) - \sum \frac{|D_v|}{|D|} \times \mathbf{Entropy}(\boldsymbol{D}_v)$$

$$\boxed{\mathbf{IG}(\mathbf{D}, \mathbf{credit\_rating}) = 0.918 - \left(\frac{|5|}{|6|} \times \mathbf{0.721} + \frac{|1|}{|6|} \times \mathbf{0}\right) = 0.317}$$

We have,

$$\mathbf{IG}(\mathbf{D}, \mathbf{Age}) = 0.918$$

$$\mathbf{IG}(\mathbf{D}, \mathbf{Income}) = 0.251$$

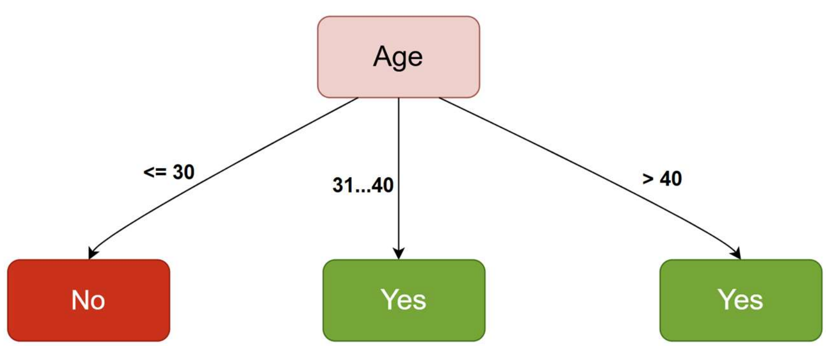
$$\mathbf{IG}(\mathbf{D}, \mathbf{Student}) = 0.109$$

$$\mathbf{IG}(\mathbf{D}, \mathbf{credit\_rating}) = 0.317$$

We Know,

$$\boldsymbol{Root\ attribut} = \boldsymbol{Attribute\ with\ Highest\ Information\ Gain}$$

$$\boldsymbol{Root\ attribut} = \boldsymbol{age}$$

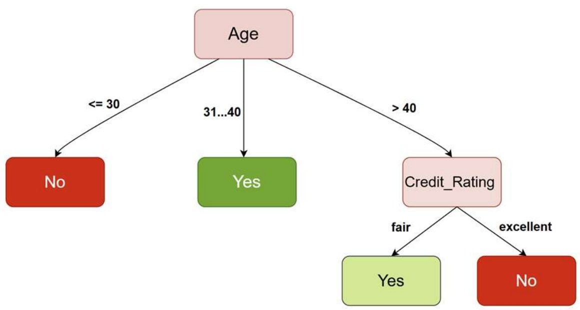


(c)

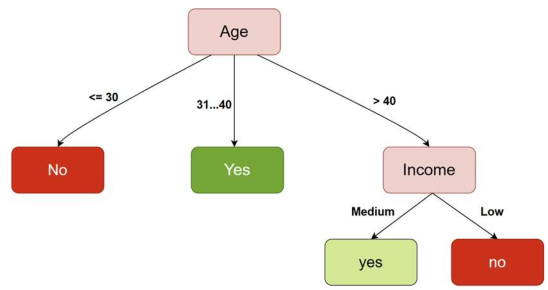
Is given new data:

Age = >40
Income = Low
Student = Yes
Credit_Rating = Fair

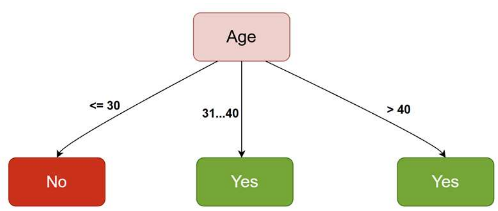
So Tress are:



Yes



No



Yes

Bagging Prediction: Bagging aggregates predictions by majority vote across the trees. With two predicting "Yes" and one predicting "No", the final class is Yes. This reduces variance compared to a single tree.

Problem2: You are given the following dataset used to predict whether a customer **buys a computer**.

ID	Age	Income	Student	Credit_Rating	Buys_Computer
1	≤30	High	No	Fair	No
2	≤30	High	No	Excellent	No
3	31– 40	High	No	Fair	Yes
4	>40	Medium	No	Fair	Yes
5	>40	Low	Yes	Fair	Yes
6	>40	Low	Yes	Excellent	No

a) Create four bootstrap samples**, each of size** 6**, from the dataset using** sampling with replacement**.**

b) Train a Random Forest classifier **using these samples**.

c) For a new customer with the following attributes:

Age = >40
Income = Low
Student = No
Credit_Rating = Fair

Predict the class (Buys_Computer) using the Random Forest model.

(a)

Sample1: ID: 1,1,3,4,5,6

ID	Age	Income	Student	Credit Rating	Buys Computer
1	≤30	High	No	Fair	No
1	≤30	High	No	Fair	No
3	31– 40	High	No	Fair	Yes
4	>40	Medium	No	Fair	Yes
5	>40	Low	Yes	Fair	Yes
6	>40	Low	Yes	Excellent	No

Sample2: ID: 1,2,3,4,2,6

ID	Age	Income	Student	Credit Rating	Buys Computer
1	≤30	High	No	Fair	No
2	≤30	High	No	Excellent	No
3	31– 40	High	No	Fair	Yes
4	>40	Medium	No	Fair	Yes
2	≤30	High	No	Excellent	No
6	>40	Low	Yes	Excellent	No

Sample3: ID: 2,1,3,4,5,3

ID	Age	Income	Student	Credit Rating	Buys Computer
2	≤30	High	No	Excellent	No
1	≤30	High	No	Fair	No
3	31– 40	High	No	Fair	Yes
4	>40	Medium	No	Fair	Yes
5	>40	Low	Yes	Fair	Yes
3	31– 40	High	No	Fair	Yes

Sample4: ID: 1,3,5,4,5,6

ID	Age	Income	Student	Credit Rating	Buys Computer
1	≤30	High	No	Fair	No
3	31– 40	High	No	Fair	Yes
5	>40	Low	Yes	Fair	Yes
4	>40	Medium	No	Fair	Yes
5	>40	Low	Yes	Fair	Yes
6	>40	Low	Yes	Excellent	No

(b)

- **For Bootstrap Sample1:** ID: 1,1,3,4,5,6
- For 1st split let consider only **age & Income** variable

We Know,

$$\boldsymbol{Entropy(D) = -p_1 log_2(p_1) - p_2 log_2(p_2)}$$

$$\boldsymbol{Entropy(D)} = -\frac{3}{6} log_2\left(\frac{3}{6}\right) - \frac{3}{6} log_2\left(\frac{3}{6}\right)$$

$$\boldsymbol{Entropy(D)} = 1$$

∴ **Entropy of the entire dataset** for the target attribute ***buys_computer***, $\boldsymbol{D} = 1$

- **For Age** (values: <=30, 31..40, >40):

Is given, For Age $\leq$ **30**:

$$\boldsymbol{Entropy(D_{v\leq 30})} = -\frac{2}{2} log_2\left(\frac{2}{2}\right) - \frac{0}{2} log_2\left(\frac{0}{2}\right)$$

$$\boldsymbol{Entropy(D_{v\leq 30})} = 0$$

Is given, For Age **31 ... 40**:

$$\boldsymbol{Entropy(D_{v\,31..40})} = -\frac{1}{1} log_2\left(\frac{1}{1}\right) - \frac{0}{1} log_2\left(\frac{0}{1}\right)$$

$$\boldsymbol{Entropy(D_{v\,31..40})} = 0$$

Is given, For Age $>$ **40**:

$$\boldsymbol{Entropy(D_{v>40})} = -\frac{2}{3} log_2\left(\frac{2}{3}\right) - \frac{1}{3} log_2\left(\frac{1}{3}\right)$$

$$\boldsymbol{Entropy(D_{v>40})} = 0.918$$

So,

$$\therefore \mathbf{IG(D, Age) = Entropy(D)} - \sum \frac{|\boldsymbol{D_v}|}{|\boldsymbol{D}|} \times \mathbf{Entropy}(\boldsymbol{D_v})$$

$$\boxed{\mathbf{IG(D, Age)} = 1 - \left(\frac{|2|}{|6|} \times \mathbf{0} + \frac{|1|}{|6|} \times \mathbf{0} + \frac{|3|}{|6|} \times \mathbf{0.918}\right) = 0.541}$$

- **For Income** (values: high, medium, low):

Is given, For income, High:

$$Entropy(D_{v\,high}) = -\frac{2}{3}log_2\left(\frac{2}{3}\right) - \frac{1}{3}log_2\left(\frac{1}{3}\right)$$

$$Entropy(D_{v\,high}) = 0.918$$

Is given, For income, Medium:

$$Entropy(D_{v\,medium}) = -\frac{1}{1}log_2\left(\frac{1}{1}\right) - \frac{0}{1}log_2\left(\frac{0}{1}\right)$$

$$Entropy(D_{v\,medium}) = 0$$

Is given, For income, Low:

$$Entropy(D_{v\,low}) = -\frac{1}{2}log_2\left(\frac{1}{2}\right) - \frac{1}{2}log_2\left(\frac{1}{2}\right)$$

$$Entropy(D_{v\,low}) = 1$$

So,

$$\therefore \mathbf{IG(D, Income\ ) = Entropy(D)} - \sum \frac{|D_v|}{|D|} \times \mathbf{Entropy}(D_v)$$

$$\boxed{\mathbf{IG(D, Income\ )} = 1 - \left(\frac{|3|}{|6|} \times \mathbf{0.918} + \frac{|1|}{|6|} \times \mathbf{0} + \frac{|2|}{|6|} \times \mathbf{1}\right) = 0.208}$$

We have,

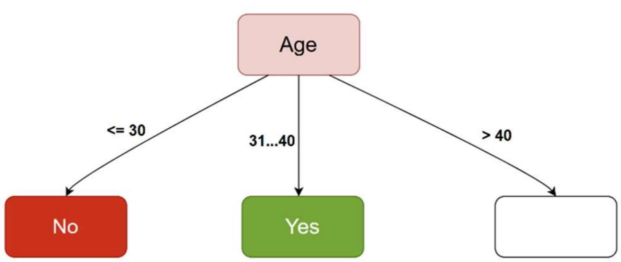
$$\mathbf{IG(D, Age)} = 0.541$$

$$\mathbf{IG(D, Income)} = 0.208$$

We Know,

$$Root\ attribut = Attribute\ with\ Highest\ Information\ Gain$$

$$Root\ attribut = age$$



Age	Income	Student	Credit Rating	Buys Computer
>40	Medium	No	Fair	Yes
>40	Low	Yes	Fair	Yes
>40	Low	Yes	Excellent	No

⇨ For 2nd split let consider only **income & Credit_Rating** variable

We Know,

$$\boldsymbol{Entropy(D) = -p_1 log_2(p_1) - p_2 log_2(p_2)}$$

$$\boldsymbol{Entropy(D)} = -\frac{2}{3}log_2\left(\frac{2}{3}\right) - \frac{1}{3}log_2\left(\frac{1}{3}\right)$$

$$\boldsymbol{Entropy(D)} = 0.918$$

∴ **Entropy of the entire dataset** for the target attribute ***buys_computer***, $\boldsymbol{D} = 0.918$

❖ **For Income** (values: high, medium, low):

Is given, For income, High:

$$\boldsymbol{Entropy(D_{v\,high})} = -\frac{0}{0}log_2\left(\frac{0}{0}\right) - \frac{0}{0}log_2\left(\frac{0}{0}\right)$$

$$\boldsymbol{Entropy(D_{v\,high})} = 0$$

Is given, For income, Medium:

$$\boldsymbol{Entropy(D_{v\,medium})} = -\frac{1}{1}log_2\left(\frac{1}{1}\right) - \frac{0}{1}log_2\left(\frac{0}{1}\right)$$

$$\boldsymbol{Entropy(D_{v\,medium})} = 0$$

Is given, For income, Low:

$$\boldsymbol{Entropy(D_{v\,low})} = -\frac{1}{2}log_2\left(\frac{1}{2}\right) - \frac{1}{2}log_2\left(\frac{1}{2}\right)$$

$$\boldsymbol{Entropy(D_{v\,low})} = 1$$

So,

$$\therefore \mathbf{IG(D, Income\ ) = Entropy(D)} - \sum \frac{|\boldsymbol{D_v}|}{|\boldsymbol{D}|} \times \mathbf{Entropy}(\boldsymbol{D_v})$$

$$\boxed{\mathbf{IG(D, Income\ )} = 0.918 - \left(\frac{|0|}{|3|} \times \mathbf{0} + \frac{|1|}{|3|} \times \mathbf{0} + \frac{|2|}{|3|} \times \mathbf{1}\right) = 0.251}$$

❖ **For credit_rating** (values: fair, excellent):

Is given, For credit_rating, fair:

$$\boldsymbol{Entropy(D_{v\,fair})} = -\frac{2}{2}log_2\left(\frac{2}{2}\right) - \frac{0}{2}log_2\left(\frac{0}{2}\right)$$

$$\boldsymbol{Entropy(D_{v\,fair})} = 0$$

Is given, For credit_rating, excellent:

$$\boldsymbol{Entropy(D_{v\,excellent})} = -\frac{1}{1}log_2\left(\frac{1}{1}\right) - \frac{0}{1}log_2\left(\frac{0}{1}\right)$$

$$\boldsymbol{Entropy(D_{v\,excellent})} = 0$$

So,

$$\therefore \mathbf{IG(D, credit\_rating)} = \mathbf{Entropy(D)} - \sum \frac{|D_v|}{|D|} \times \mathbf{Entropy}(D_v)$$

$$\boxed{\mathbf{IG(D, credit\_rating)} = 0.918 - \left(\frac{|2|}{|3|} \times \mathbf{0} + \frac{|1|}{|3|} \times \mathbf{0}\right) = 0.918}$$

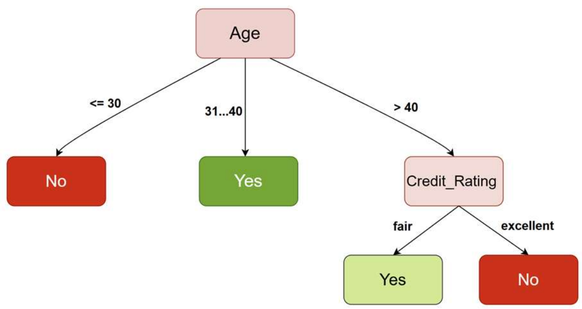
We have,

$$\mathbf{IG(D, Income)} = 0.251$$
$$\mathbf{IG(D, credit\_rating\ )} = 0.918$$

We Know,

$$\boldsymbol{Root\ attribut = Attribute\ with\ Highest\ Information\ Gain}$$

$$\boldsymbol{Root\ attribut} = \mathbf{credit\_rating}$$



- **For Bootstrap Sample2:** ID: 1,2,3,4,2,6
- For 1st split let consider only **Student & Income** variable

We Know,

$$\boldsymbol{Entropy(D) = -p_1 log_2(p_1) - p_2 log_2(p_2)}$$

$$\boldsymbol{Entropy(D)} = -\frac{4}{6} log_2\left(\frac{4}{6}\right) - \frac{2}{6} log_2\left(\frac{2}{6}\right)$$

$$\boldsymbol{Entropy(D)} = 0.918$$

$\therefore$ **Entropy of the entire dataset** for the target attribute ***buys_computer***, $\boldsymbol{D} = 0.918$

- **For Income** (values: high, medium, low):

  Is given, For income, High:

$$\boldsymbol{Entropy(D_{v\ high})} = -\frac{3}{4} log_2\left(\frac{3}{4}\right) - \frac{1}{4} log_2\left(\frac{1}{4}\right)$$

$$\boldsymbol{Entropy(D_{v\ high})} = 0.811$$

Is given, For income, Medium:

$$\boldsymbol{Entropy}(\boldsymbol{D}_{\boldsymbol{v}\ \boldsymbol{medium}}) = -\frac{1}{1}log_2\left(\frac{1}{1}\right) - \frac{0}{1}log_2\left(\frac{0}{1}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D}_{\boldsymbol{v}\ \boldsymbol{medium}}) = 0$$

Is given, For income, Low:

$$\boldsymbol{Entropy}(\boldsymbol{D}_{\boldsymbol{v}\ \boldsymbol{low}}) = -\frac{1}{1}log_2\left(\frac{1}{1}\right) - \frac{0}{1}log_2\left(\frac{0}{1}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D}_{\boldsymbol{v}\ \boldsymbol{low}}) = 0$$

So,

$$\therefore \mathbf{IG}(\mathbf{D}, \mathbf{Income}\ ) = \mathbf{Entropy}(\mathbf{D}) - \sum \frac{|\boldsymbol{D}_{\boldsymbol{v}}|}{|\boldsymbol{D}|} \times \mathbf{Entropy}(\boldsymbol{D}_{\boldsymbol{v}})$$

$$\boxed{\mathbf{IG}(\mathbf{D}, \mathbf{Income}\ ) = 0.918 - \left(\frac{|4|}{|6|} \times \mathbf{0.811} + \frac{|1|}{|6|} \times \mathbf{0} + \frac{|1|}{|6|} \times \mathbf{0}\right) = 0.377}$$

❖ **For Student** (values: yes, no):

Is given, For Student, yes:

$$\boldsymbol{Entropy}(\boldsymbol{D}_{\boldsymbol{v}\ \boldsymbol{yes}}) = -\frac{1}{1}log_2\left(\frac{1}{1}\right) - \frac{0}{1}log_2\left(\frac{0}{1}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D}_{\boldsymbol{v}\ \boldsymbol{yes}}) = \mathbf{0}$$

Is given, For Student, No:

$$\boldsymbol{Entropy}(\boldsymbol{D}_{\boldsymbol{v}\ \boldsymbol{no}}) = -\frac{3}{5}log_2\left(\frac{3}{5}\right) - \frac{2}{5}log_2\left(\frac{2}{5}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D}_{\boldsymbol{v}\ \boldsymbol{no}}) = \mathbf{0}.970$$

So,

$$\therefore \mathbf{IG}(\mathbf{D}, \mathbf{Student}\ ) = \mathbf{Entropy}(\mathbf{D}) - \sum \frac{|\boldsymbol{D}_{\boldsymbol{v}}|}{|\boldsymbol{D}|} \times \mathbf{Entropy}(\boldsymbol{D}_{\boldsymbol{v}})$$

$$\boxed{\mathbf{IG}(\mathbf{D}, \mathbf{Student}\ ) = 0.918 - \left(\frac{|5|}{|6|} \times \mathbf{0.97} + \frac{|1|}{|6|} \times \mathbf{0}\right) = 0.109}$$

We have,

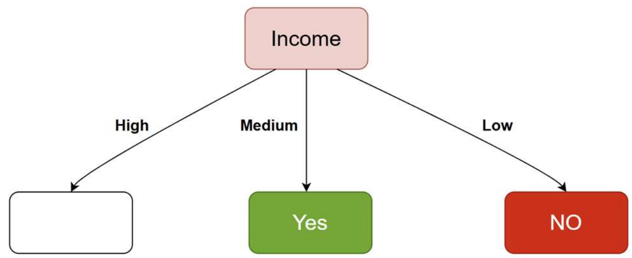
$$\mathbf{IG}(\mathbf{D}, \mathbf{Income}) = 0.377$$

$$\mathbf{IG}(\mathbf{D}, \mathbf{Student}) = 0.109$$

We Know,

$$Root\ attribut = Attribute\ with\ Highest\ Information\ Gain$$

$$\boldsymbol{Root\ attribut = Income}$$



ID	Age	Income	Student	Credit Rating	Buys Computer
1	≤30	High	No	Fair	No
2	≤30	High	No	Excellent	No
3	31– 40	High	No	Fair	Yes
2	≤30	High	No	Excellent	No

⇨

⇨ For 2nd split let consider only **Age & Credit_Rating** variable

We Know,

$$\boldsymbol{Entropy(D) = -p_1 log_2(p_1) - p_2 log_2(p_2)}$$

$$\boldsymbol{Entropy(D)} = -\frac{1}{4} log_2\left(\frac{1}{4}\right) - \frac{3}{4} log_2\left(\frac{3}{4}\right)$$

$$\boldsymbol{Entropy(D)} = 0.811$$

∴ **Entropy of the entire dataset** for the target attribute ***buys_computer***, $\boldsymbol{D} = 0.811$

❖ **For Age** (values: <=30, 31..40, >40):

Is given, For Age $\le$ **30**:

$$\boldsymbol{Entropy(D_{v\le 30})} = -\frac{3}{3} log_2\left(\frac{3}{3}\right) - \frac{0}{3} log_2\left(\frac{0}{3}\right)$$

$$\boldsymbol{Entropy(D_{v\le 30})} = 0$$

Is given, For Age **31 ... 40**:

$$\boldsymbol{Entropy(D_{v\,31..40})} = -\frac{1}{1} log_2\left(\frac{1}{1}\right) - \frac{0}{1} log_2\left(\frac{0}{1}\right)$$

$$\boldsymbol{Entropy(D_{v\,31..40})} = 0$$

So,

$$\therefore \mathbf{IG(D, Age)} = \mathbf{Entropy(D)} - \sum \frac{|D_v|}{|D|} \times \mathbf{Entropy}(D_v)$$

$$\boxed{\mathbf{IG(D, Age)} = 0.811 - \left(\frac{|3|}{|4|} \times \mathbf{0} + \frac{|1|}{|4|} \times \mathbf{0}\right) = 0.811}$$

- **For credit_rating** (values: fair, excellent):

Is given, For credit_rating, fair:

$$Entropy(D_{v\,fair}) = -\frac{1}{2} log_2\left(\frac{1}{2}\right) - \frac{1}{2} log_2\left(\frac{1}{2}\right)$$

$$Entropy(D_{v\,fair}) = 1$$

Is given, For credit_rating, excellent:

$$Entropy(D_{v\,excellent}) = -\frac{2}{2} log_2\left(\frac{2}{2}\right) - \frac{0}{2} log_2\left(\frac{0}{2}\right)$$

$$Entropy(D_{v\,excellent}) = 0$$

So,

$$\therefore \mathbf{IG(D, credit\_rating)} = \mathbf{Entropy(D)} - \sum \frac{|D_v|}{|D|} \times \mathbf{Entropy}(D_v)$$

$$\boxed{\mathbf{IG(D, credit\_rating)} = 0.811 - \left(\frac{|2|}{|4|} \times \mathbf{1} + \frac{|2|}{|4|} \times \mathbf{0}\right) = 0.311}$$

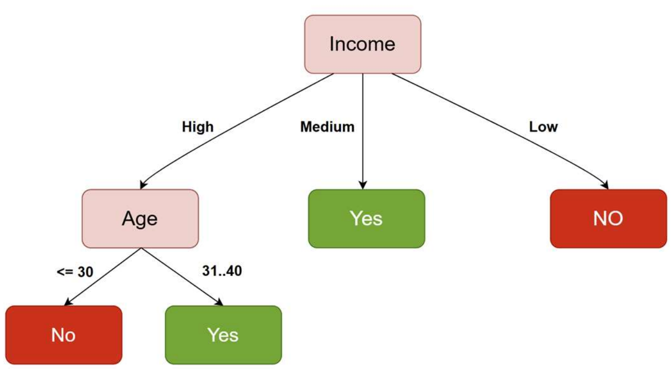
We have,

$$\mathbf{IG(D, Age)} = 0.811$$
$$\mathbf{IG(D, credit\_rating\ )} = 0.311$$

We Know,

$$Root\ attribut = Attribute\ with\ Highest\ Information\ Gain$$

$$Root\ attribut = \mathbf{Age}$$



⇨ **For Bootstrap Sample3:** ID: 2,1,3,4,5,3

⇨ For 1st split let consider only **age & Student** variable

We Know,

$$\boldsymbol{Entropy}(\boldsymbol{D}) = -\boldsymbol{p_1 log_2}(\boldsymbol{p_1}) - \boldsymbol{p_2 log_2}(\boldsymbol{p_2})$$

$$\boldsymbol{Entropy}(\boldsymbol{D}) = -\frac{4}{6} log_2\left(\frac{4}{6}\right) - \frac{2}{6} log_2\left(\frac{2}{6}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D}) = 0.918$$

∴ **Entropy of the entire dataset** for the target attribute ***buys_computer***, $\boldsymbol{D} = 0.918$

❖ **For Age** (values: <=30, 31..40, >40):

Is given, For Age $\leq$ **30**:

$$\boldsymbol{Entropy}(\boldsymbol{D_{v\leq 30}}) = -\frac{2}{2} log_2\left(\frac{2}{2}\right) - \frac{0}{2} log_2\left(\frac{0}{2}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D_{v\leq 30}}) = 0$$

Is given, For Age **31 ... 40**:

$$\boldsymbol{Entropy}(\boldsymbol{D_{v\,31..40}}) = -\frac{2}{2} log_2\left(\frac{2}{2}\right) - \frac{0}{2} log_2\left(\frac{0}{2}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D_{v\,31..40}}) = 0$$

Is given, For Age $>$ **40**:

$$\boldsymbol{Entropy}(\boldsymbol{D_{v>40}}) = -\frac{2}{2} log_2\left(\frac{2}{2}\right) - \frac{0}{2} log_2\left(\frac{0}{2}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D_{v>40}}) = 0$$

So,

$$\therefore \mathbf{IG}(\mathbf{D}, \mathbf{Age}) = \mathbf{Entropy}(\mathbf{D}) - \sum \frac{|\boldsymbol{D_v}|}{|\boldsymbol{D}|} \times \mathbf{Entropy}(\boldsymbol{D_v})$$

$$\boxed{\mathbf{IG}(\mathbf{D}, \mathbf{Age}) = 0.918 - \left(\frac{|2|}{|6|} \times \mathbf{0} + \frac{|2|}{|6|} \times \mathbf{0} + \frac{|2|}{|6|} \times \mathbf{0}\right) = 0.918}$$

- **For Student** (values: yes, no):

Is given, For Student, yes:

$$\boldsymbol{Entropy}(\boldsymbol{D}_{v\,yes}) = -\frac{1}{1}log_2\left(\frac{1}{1}\right) - \frac{0}{1}log_2\left(\frac{0}{1}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D}_{v\,yes}) = \mathbf{0}$$

Is given, For Student, No:

$$\boldsymbol{Entropy}(\boldsymbol{D}_{v\,no}) = -\frac{3}{5}log_2\left(\frac{3}{5}\right) - \frac{2}{5}log_2\left(\frac{2}{5}\right)$$

$$\boldsymbol{Entropy}(\boldsymbol{D}_{v\,no}) = \mathbf{0.}970$$

So,

$$\therefore \mathbf{IG}(\mathbf{D}, \mathbf{Student}) = \mathbf{Entropy}(\mathbf{D}) - \sum \frac{|D_v|}{|D|} \times \mathbf{Entropy}(\boldsymbol{D}_v)$$

$$\boxed{\mathbf{IG}(\mathbf{D}, \mathbf{Student}) = 0.918 - \left(\frac{|5|}{|6|} \times \mathbf{0.97} + \frac{|1|}{|6|} \times \mathbf{0}\right) = 0.109}$$

We have,

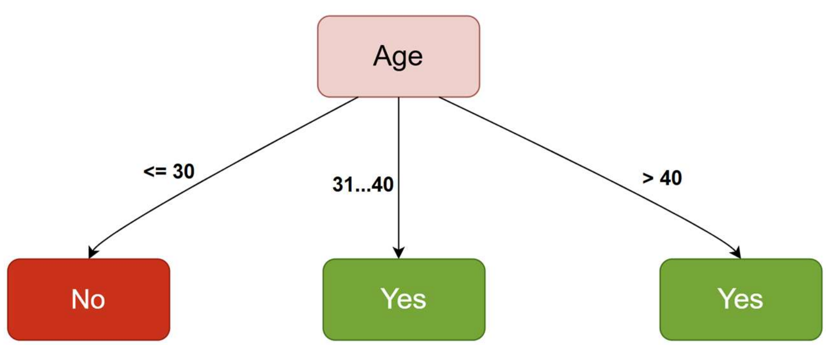
$$\mathbf{IG}(\mathbf{D}, \mathbf{Age}) = 0.918$$

$$\mathbf{IG}(\mathbf{D}, \mathbf{Student}) = 0.109$$

We Know,

$$\boldsymbol{Root\ attribut = Attribute\ with\ Highest\ Information\ Gain}$$

$$\boldsymbol{Root\ attribut = age}$$

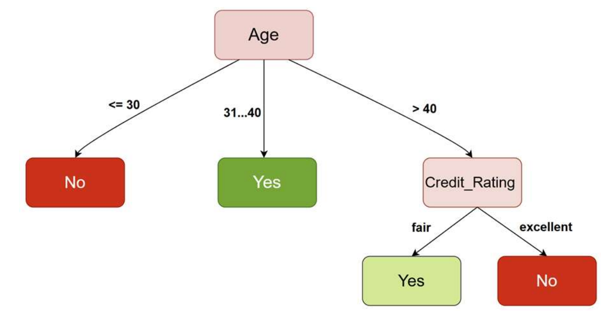


(c)

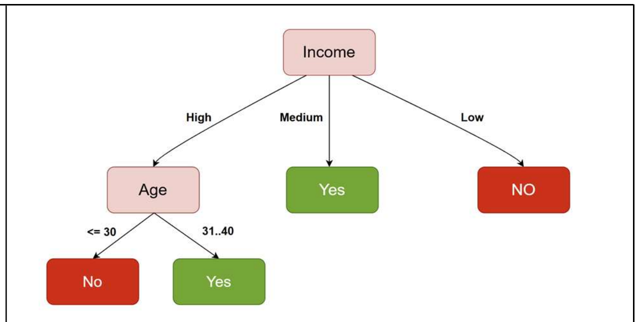
Is given new data:

Age = >40
Income = Low
Student = No
Credit_Rating = Fair

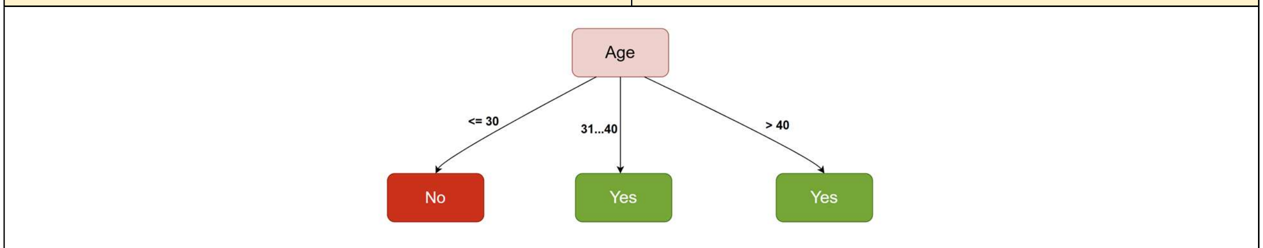
So Tress are:



Yes



No



Yes

Random Forest Prediction: Random Forest aggregates predictions by majority vote across the trees. With two predicting "Yes" and one predicting "No", the final class is Yes. This reduces variance compared to a single tree.

PERFORMANCE MEASURE IN CLASSIFICATION

Performance Measures in Classification

⇨ After training a machine learning model, its effectiveness is evaluated using a **test dataset** that contains unseen examples. Good performance on test data indicates that the model **generalizes well**. Various **formal evaluation metrics** are used to assess classification models.

❖ Common Classification Performance Metrics

- **Confusion Matrix**
- **Accuracy**
- **Cost-Sensitive Accuracy**
- **Precision**
- **Recall**
- **F1- Score**
- **ROC Curve and AUC**

Confusion Matrix

⇨ A **confusion matrix** is a table that summarizes how well a classification model performs by comparing **actual labels** with **predicted labels**.

❖ Binary Classification Confusion Matrix:

⇨ For two classes (e.g., *spam* and *not spam*):

- **True Positive (TP):** Correctly predicted positive cases
- **True Negative (TN):** Correctly predicted negative cases
- **False Positive (FP):** Incorrectly predicted positive cases
- **False Negative (FN):** Incorrectly predicted negative cases

❖ Example:

	Spam (predicted)	Not_spam (predicted)
Spam (Actual)	23 (TP)	1 (FN)
Not_spam (Actual)	12 (FP)	556 (TN)

- ➢ TP = 23 (spam correctly identified)
- ➢ FN = 1 (spam predicted as not spam)
- ➢ TN = 556 (not spam correctly identified)
- ➢ FP = 12 (not spam predicted as spam)

Multiclass Confusion Matrix

⇨ In **multiclass classification**, the confusion matrix contains **multiple rows and columns**, one for each class.

	Predicted: A	**Predicted: B**	**Predicted: C**
Actual: A	5	2	1
Actual: B	1	4	2
Actual: C	0	1	6

- **Diagonal elements:** Correct predictions
- **Off-diagonal elements:** Misclassifications
- Helps identify **error patterns** between classes

❖ **Use Case:**

⇨ A confusion matrix could reveal that a model trained to recognize different species of animals tends to mistakenly predict "cat" instead of "panther," or "mouse" instead of "rat.":

- In this case, you can decide to **add more labeled examples** of these species to help the learning algorithm to "see" the difference between them
- Alternatively, you might **add additional features** the learning algorithm can use to build a model that would better distinguish between these species.

Precision

⇨ **Precision** answers the question: *Of all predicted positive instances, how many are actually positive?*

$$\textbf{Precision} = \frac{\textbf{TP}}{\textbf{TP} + \textbf{FP}}$$

❖ **Example:** If 100 patients are predicted to have a disease and only 80 actually have it:

$$\text{Precision} = \frac{100}{80} = 0.8 = 80\%$$

Recall

⇨ **Recall** answers the question: *Of all actual positive instances, how many did the model correctly identify?*

$$\boxed{\mathbf{Recall} = \frac{\mathbf{TP}}{\mathbf{TP + FN}}}$$

❖ **Example:** If 90 people actually have a disease and 80 are correctly identified:

$$\text{Recall} = \frac{80}{90} = 0.89 = 89\%$$

Precision vs Recall

Metric	Focus	Important When
Precision	Correctness of positive predictions	False positives are costly (e.g., spam detection)
Recall	Capturing all positives	Missing positives is costly (e.g., cancer detection)

F1 Score

⇨ The **F1 Score** is a single metric that balances both **precision and recall**.

$$\boxed{\mathbf{F1\ Score} = \frac{\mathbf{2 \times Precision\ \times Recall}}{\mathbf{Precision + Recall}}}$$

❖ **When F1-Score Is Useful:**

- F1 Score is high only when **both precision and recall are high**
- Useful when false positives and false negatives are both important
- Commonly used in medical diagnosis, fraud detection, and spam filtering

Accuracy

⇨ **Accuracy** measures the overall correctness of the classification model.

⇨ Accuracy is the ratio of correctly classified instances to the total number of instances.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

❖ When Accuracy Is Useful:

- When all types of errors are **equally important**
- When class distribution is **balanced**

Choice of Metric and Trade-offs

⇨ The choice of evaluation metric depends on:

- **Cost**
- **Risk**
- **Application domain**

❖ Example (Spam Detection):

- High recall → Catch most spam emails
- High precision → Avoid misclassifying important emails
- Often a balance is chosen based on practical needs

Metric	When to Use
Accuracy	For balanced datasets; use only as a rough indicator and with other metrics
Recall (TPR)	When false negatives are more costly
False Positive Rate	When false positives are more costly
Precision	When positive predictions must be highly accurate

ROC Curve and AUC

❖ ROC (Receiver Operating Characteristic) Curve

- Plots **True Positive Rate (TPR)** vs **False Positive Rate (FPR)**
- Generated by varying the **classification threshold**

❖ AUC (Area Under the Curve)

- Measures overall classifier performance
- **AUC > 0.5:** Better than random guessing
- **AUC = 1.0:** Perfect classifier
- **AUC < 0.5:** Model is incorrect or flawed

Training Loss

⇨ **Training Loss** measures the error between the model's predictions and actual labels on the **training dataset**.

❖ How Training Loss Works

1. Predictions are made during training
2. Error is calculated using a **loss function**
3. Optimization algorithms (e.g., gradient descent) minimize this loss

❖ Expected Behavior

- Training loss should **decrease over time**
- Low training loss + high validation loss → **Overfitting**

❖ Common Loss Functions

- Mean Squared Error (MSE) – Regression
- Cross-Entropy Loss – Classification
- Hinge Loss – Support Vector Machines

Parameters vs Hyperparameters

❖ Parameters

- Learned automatically from training data
- Examples:
  - Linear regression: weights, intercept
  - Neural networks: weights, biases

❖ Hyperparameters

- Set **before training**
- Control learning behavior and model complexity

❖ Examples:

- Learning rate
- Number of hidden layers
- Batch size
- Tree depth

❖ Key Difference:

- Parameters → learned
- Hyperparameters → manually set and tuned

Hyperparameter Tuning

⇨ **Hyperparameter tuning** is the process of finding the optimal hyperparameter values to improve model performance

❖ Common Methods

1. Grid Search
2. Random Search

Grid Search

- Exhaustively evaluates **all combinations** of hyperparameters
- Ensures optimal coverage
- Computationally expensive

❖ Example:

- $a_1 = [0,1,2,3,4,5]$
- $a_2 = [10,20,30,40,50,60]$
- $a_3 = [105,110,115,120,125]$

All possible combinations are tested.

Random Search

- Samples hyperparameter values from **probability distributions**
- Faster and more efficient
- Often finds good solutions with fewer iterations

❖ Why Random Search Works Better

- Not all hyperparameters equally affect performance
- Random sampling explores important regions faster